

MARYBOROUGH, Australia

WMO: 945670

Lat: 25.5161S

Lon: 152.7156E

Elev: 35

StdP: 14.68

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	41.4	44.3	29.8	23.9	62.9	33.3	27.9	58.9	20.7	66.1	18.7	66.1	2.3	230	0.350

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.2	90.9	74.8	88.8	74.0	87.0	73.2	78.3	85.6	77.2	84.0	76.1	82.7	10.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.5	138.2	81.6	75.3	132.7	80.4	74.2	127.7	79.6	41.9	85.9	40.7	84.0	39.7	82.8	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.9	18.8	16.9	35.2	95.7	1.9	2.2	33.9	97.2	32.8	98.5	31.7	99.8	30.3	101.3		
(5)			34.1	81.3	1.9	2.3	32.7	83.0	31.6	84.3	30.6	85.6	29.2	87.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	69.8	78.0	77.6	75.8	71.3	65.8	61.6	60.0	61.4	66.4	70.2	73.9	76.6	(6)
(7)		DBStd	7.32	3.07	2.99	2.71	2.98	3.93	4.20	3.87	3.96	3.81	3.64	3.58	3.34	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	466	0	0	0	2	37	115	159	125	26	2	0	0	(9)
(10)		CDD50	7240	869	774	800	638	489	347	311	353	491	627	716	825	(10)
(11)		CDD65	2232	404	354	335	190	62	12	5	13	67	164	266	360	(11)
(12)		CDH74	15657	3320	2730	2141	866	207	30	32	102	474	995	1864	2896	(12)
(13)		CDH80	4365	1113	881	560	115	9	0	1	3	52	173	507	951	(13)
(14)	Wind	WSAvg	7.0	8.2	8.3	7.8	7.1	6.0	5.9	5.7	6.0	6.7	7.3	7.5	7.9	(14)
(15)	Precipitation	PrecAvg	42.1	5.9	6.7	4.8	3.0	3.2	2.2	1.9	1.6	1.3	3.0	3.8	4.9	(15)
(16)		PrecMax	65.7	19.8	35.4	20.4	11.5	13.5	13.7	16.0	4.3	5.2	8.5	11.9	12.8	(16)
(17)		PrecMin	24.3	0.2	0.5	0.6	0.7	0.1	0.0	0.0	0.0	0.0	0.3	0.4	0.7	(17)
(18)		PrecStd	10.7	4.1	5.9	3.7	2.4	2.3	2.5	2.5	1.0	1.3	1.7	2.4	2.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.2	92.8	90.5	85.5	81.1	77.3	77.1	79.7	84.4	87.6	90.0	93.2	(19)
(20)			MCWB	76.6	75.5	74.7	70.8	67.4	65.6	62.2	63.7	64.7	68.3	71.4	74.1	(20)
(21)		2%	DB	90.4	90.0	87.5	83.1	78.2	74.7	74.3	76.8	81.4	83.8	87.2	89.8	(21)
(22)			MCWB	75.6	75.7	73.8	69.9	65.9	63.8	61.3	62.4	65.1	68.5	71.4	73.9	(22)
(23)		5%	DB	88.0	87.5	85.3	81.0	76.3	72.8	71.9	74.5	79.0	81.7	85.2	87.4	(23)
(24)			MCWB	74.9	74.6	72.7	68.9	65.0	62.8	60.2	61.0	64.3	67.9	70.7	73.1	(24)
(25)		10%	DB	85.8	85.3	83.1	79.1	74.3	70.6	69.8	72.2	76.8	79.6	83.0	85.3	(25)
(26)			MCWB	73.5	73.6	71.9	68.3	64.2	61.7	59.5	59.8	63.8	66.8	69.5	72.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.6	80.0	78.1	75.2	71.9	68.4	67.6	68.3	71.1	73.8	76.5	79.1	(27)
(28)			MCDWB	88.2	88.4	83.7	81.6	76.1	73.2	72.6	75.4	77.9	82.5	85.4	87.0	(28)
(29)		2%	WB	78.3	78.0	76.4	72.7	69.9	66.5	65.2	66.1	69.4	72.0	74.9	77.2	(29)
(30)			MCDWB	85.4	85.6	82.3	78.7	73.7	71.1	70.1	71.9	76.6	79.0	82.5	84.5	(30)
(31)		5%	WB	77.1	76.8	75.2	71.3	68.0	65.0	63.1	64.4	67.8	70.5	73.4	75.7	(31)
(32)			MCDWB	83.6	83.8	81.4	77.0	73.2	70.0	68.4	70.3	74.3	77.2	80.2	82.5	(32)
(33)		10%	WB	75.9	75.6	74.0	70.1	66.4	63.5	61.5	62.7	66.4	69.1	71.9	74.4	(33)
(34)			MCDWB	82.3	81.9	80.1	76.1	71.8	68.6	67.0	68.5	72.7	75.7	78.8	81.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.2	14.7	15.2	16.4	18.2	19.0	21.5	23.2	22.0	18.8	17.6	16.5	(35)
(36)			MCDBR	18.5	18.2	18.6	19.1	20.3	20.2	24.0	25.9	25.0	21.7	20.6	19.6	(36)
(37)			MCWBR	6.1	6.0	6.6	7.7	9.4	10.4	12.1	12.6	10.8	8.4	7.4	6.6	(37)
(38)		5% WB	MCDWB	15.0	15.1	15.0	15.5	15.1	16.1	18.0	20.0	18.8	17.2	16.3	15.7	(38)
(39)			MCWBR	6.1	6.1	6.3	7.2	8.1	9.1	10.6	11.4	9.1	7.6	7.3	6.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.391	0.385	0.368	0.343	0.323	0.317	0.307	0.318	0.356	0.380	0.378	0.394	(40)	
(41)		taud	2.508	2.529	2.573	2.626	2.633	2.640	2.651	2.605	2.489	2.455	2.496	2.480	(41)	
(42)		Ebn at Noon	302	299	295	289	282	277	285	293	294	297	304	302	(42)	
(43)		Edh at Noon	36	35	32	28	25	24	24	28	34	37	36	37	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2082	1886	1650	1456	1224	1044	1192	1471	1792	1959	2114	2122	(44)	
(45)		RadStd	159	162	128	109	61	79	102	112	133	165	202	229	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-0.77	-1.01	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)		+0.54	N/A	N/A	N/A	N/A	N/A	N/A	-144	+374	+236

Nomenclature: See separate page